

Prairie Telehealth Network (PTN): Vendor Data Onboarding & Integration Process

A Business Analyst Portfolio Project

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About This Project

This is a fictional case study modelled loosely on real business analysis experience gained while working on Virtual Visit, Telehealth, and Remote Patient Monitoring programs at a Canadian provincial health authority. The organization, vendors, and data shown here are entirely fictional and contain no proprietary information. The project simulates a common BA challenge: standardizing inconsistent vendor data into a unified reporting process, using a two-tier approach a standardized monthly summary for external reporting, and structured vendor detail data for internal service dashboards.

Contents

1. **Project Brief:** problem definition, scope, and objectives
2. **Current State Process Map:** the manual reconciliation process and its pain points
3. **Stakeholder Analysis:** Power/Interest Matrix and RACI Chart
4. **Requirements Document:** Business, Functional, and Non-Functional requirements
5. **Data Mapping Table:** 3 fictional vendors mapped to a standardized summary schema
6. **Future State Process Map:** the redesigned, standardized process

Tools used: Microsoft Word & Excel, Lucidchart, fictional data modelling

Prairie Telehealth Network (PTN) - BA Portfolio Project

Fictional case study

Phase 1 of 7: Project Brief

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TABLE OF CONTENTS

Background.....	2
Business Problem.....	2
Project Objective.....	2
Scope.....	3
In scope.....	3
Out of scope.....	3
Assumptions.....	3
Constraints.....	3
Success Criteria.....	3
High-Level Stakeholders.....	4
Project Phases.....	4

Project Brief:

Vendor Data Onboarding & Integration Process

Background

PTN is a fictional provincial health authority that coordinates virtual care services across urban and rural communities, including virtual visits, telehealth consultations, and remote patient monitoring (RPM). To scale these services, PTN contracts with third-party technology vendors who provide the platforms and devices patients and clinicians use.

Each vendor delivers usage and clinical data back to PTN in a format that is internally standardized for their own platform but no standardized summary layer exists across vendors. There is no shared data dictionary defining how vendor-specific fields should roll up into a common set of program metrics.

Business Problem

The PTN Analytics & Reporting team manually reconciles vendor files each month to produce program performance reports (utilization, patient outcomes, SLA compliance) for provincial health ministry stakeholders. Because each vendor's detailed data is structured differently, the same monthly Ministry metrics must be manually re-derived from three structurally different datasets each cycle. Separately, no consistent structure exists for turning vendor detail data into service-specific operational dashboards used internally by program teams.

This manual, ad hoc process is:

- **Slow:** approximately 6-8 business days per reporting cycle spent on cleanup and reconciliation rather than analysis
- **Error-prone:** mismatched field definitions (e.g., one vendor's "session_duration" is recorded in minutes, another's in seconds) have caused reporting inaccuracies
- **Not scalable:** PTN plans to onboard 2-3 new telehealth vendors within the next 12 months, and the current process cannot absorb that growth without adding headcount

Project Objective

Design and document a standardized vendor data onboarding process for PTN, including:

- A standardized monthly summary schema, extracted from each vendor's detailed data, for consistent reporting to the Ministry of Health and other external stakeholders
- A defined approach for using each vendor's detailed data to support service-specific operational dashboards (Virtual Visits, Telehealth, RPM)
- A repeatable data mapping methodology so future vendors can be onboarded with minimal manual rework

Scope

In scope

- Current state process mapping (how vendor data is received and reconciled today)
- Stakeholder analysis for the onboarding process
- Business, functional, and non-functional requirements for the summary extraction process and dashboard data usage
- Data mapping table for 3 representative fictional vendors: detailed vendor fields → standardized summary schema
- Conceptual definition of service-level dashboard data needs (not dashboard build)
- Future state process map reflecting the recommended process

Out of scope

- Actual system/ETL build or dashboard development (this is a business analysis deliverable, not a technical implementation)
- Vendor contract or legal terms
- Real-time data integration (initial scope is batch/periodic file submission)

Assumptions

- Vendors submit detailed data via secure file transfer (e.g., SFTP) on a monthly cadence
- PTN has an existing data warehouse/reporting layer that would consume both the summary and detail schemas
- Vendors have sufficient technical capability to export data against a documented specification

Constraints

- No additional budget for new tooling in year one. The solution should work within PTN's existing reporting infrastructure
- Vendors vary in technical maturity; the summary extraction specification must be usable by less sophisticated vendors, not just enterprise-grade ones

Success Criteria

- Reduce manual reconciliation effort from ~6-8 days to ~2 days per reporting cycle
- Zero data mapping errors caused by undocumented field definitions
- New vendor onboarding time reduced from ~4-6 weeks to ~2 weeks

High-Level Stakeholders

Full stakeholder matrix and RACI chart are developed in Phase 3.

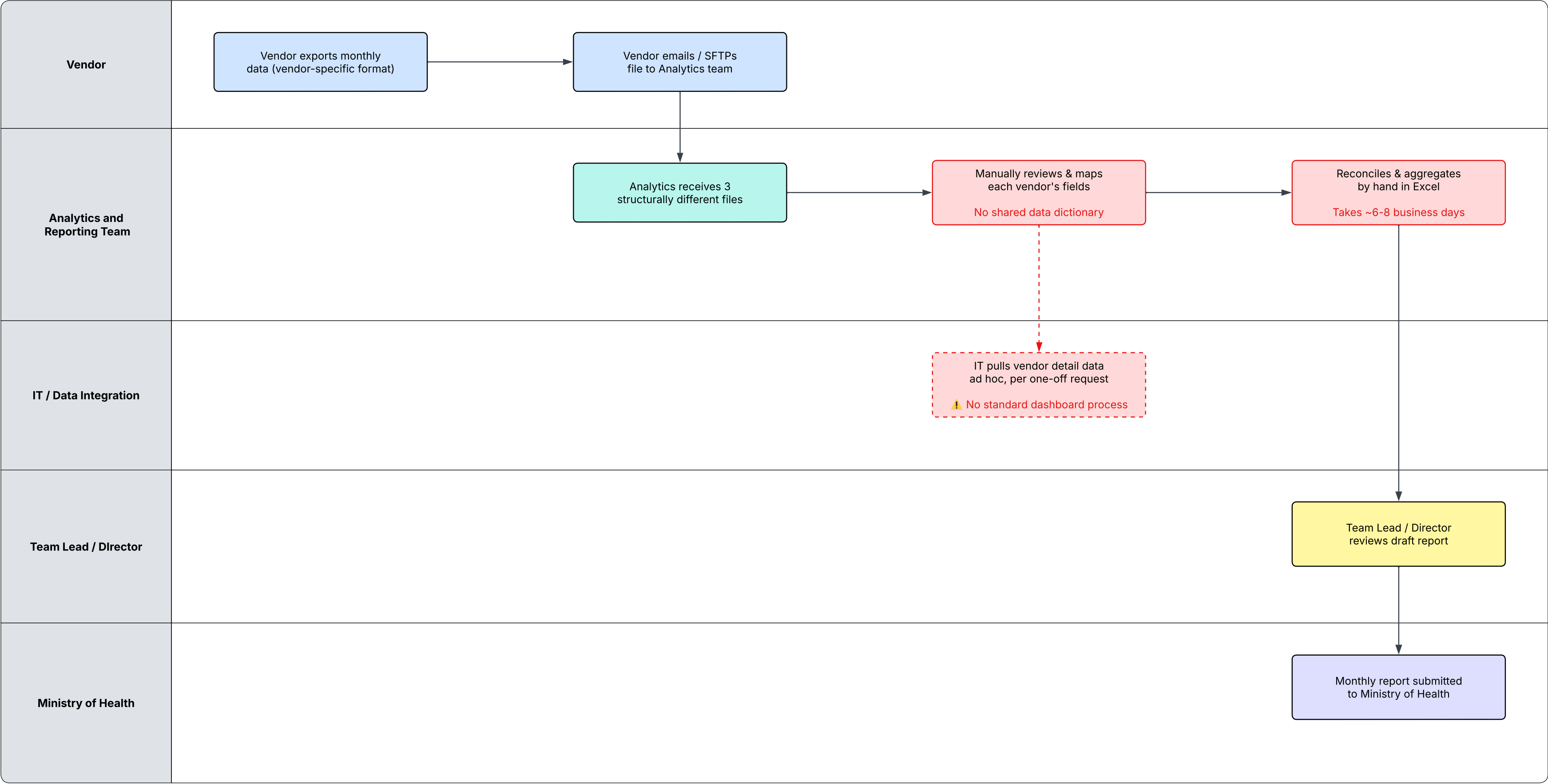
Role	Interest in this project
Director, Virtual Care Programs	Project sponsor; accountable for program reporting accuracy to the Ministry
Analytics & Reporting Team Lead	Owns the monthly reconciliation process this project aims to fix
Vendor Relationship Manager	Manages vendor communication and onboarding timelines
IT / Data Integration Lead	Will implement the summary extraction logic and dashboard data feeds
Vendor Technical Contacts (x3)	Provide detailed-format data; must adopt the summary specification
Ministry of Health Reporting Liaison	Primary consumer of the standardized monthly summary reports

Project Phases

- 1. Project Brief (this document)
- 2. Current State Process Map
- 3. Stakeholder Analysis (Matrix + RACI)
- 4. Requirements Document (Business, Functional, Non-Functional)
- 5. Data Mapping Table (3 vendors → standardized summary schema)
- 6. Future State Process Map
- 7. Final Portfolio Package

Current State: Vendor Data Reconciliation Process (PTN)

Phase 2 - Manual monthly reconciliation across 3 telehealth vendors



----- Ad hoc / unstandardized process

Pain points / bottlenecks

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Phase 3 of 7: Stakeholder Analysis
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Stakeholder Matrix

Power / Interest grid (who to engage, and how)

Stakeholder	Group	Power	Interest	Quadrant	Engagement Strategy	Key Concern
Director, Virtual Care Programs	Sponsor	High	High	Manage Closely	Involve in key milestone reviews; escalation point for scope/priority decisions	Ministry accountability and program reputation if reporting is inaccurate or late
Analytics & Reporting Team Lead	Internal Ops	High	High	Manage Closely	Partner closely throughout; primary source for current-state pain points and validation of requirements	Reducing manual reconciliation effort without losing data accuracy
IT / Data Integration Lead	Internal Ops	Medium	High	Manage Closely	Involve early in schema and mapping design so the specification is technically buildable	Feasibility of extraction rules; ongoing maintenance burden of the new process
Vendor Relationship Manager	Internal Ops	Medium	High	Manage Closely	Co-own vendor communication; ensure specification is realistic for vendors to implement	Vendor pushback or delay if new requirements are seen as burdensome
Program / Service Team Leads (Virtual Visits, Telehealth, RPM)	Internal Ops	Medium	High	Manage Closely	Consult on dashboard data needs per service line; validate detail-schema usage	Whether their service-specific metrics are represented in the new structure
Ministry of Health Reporting Liaison	External	High	Medium	Keep Satisfied	Periodic updates on timeline and reporting continuity; no day-to-day involvement needed	Uninterrupted, accurate monthly reporting during the transition
Vendor Technical Contacts (x3)	External	Low	Medium	Keep Informed	Share the summary specification and mapping requirements once finalized; light-touch check-ins	Effort required to adjust their export process; unclear or shifting requirements

Quadrant Legend

Manage Closely	High power, high interest (engage frequently, involve in decisions)
Keep Satisfied	High power, lower interest (keep informed at a summary level, avoid overloading)
Keep Informed	Lower power, high interest (communicate proactively, low decision involvement)
Monitor	Low power, low interest (minimal effort, monitor for change)

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Phase 3 of 7: RACI chart
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RACI Chart

Roles and accountability across the vendor onboarding process

Activity	Business Analyst	Director (Sponsor)	Analytics & Reporting Lead	IT / Data Integration Lead	Vendor Relationship Mgr	Program / Service Leads	Vendor Technical Contacts	Ministry Liaison
Define business & functional requirements	R	A	C	C	C	C	I	I
Design standardized summary schema	R	I	A	C	I	I	I	C
Define dashboard data requirements (detail schema usage)	R	I	C	C	I	A	I	I
Build vendor-to-summary data mapping specification	R	I	A	C	C	I	C	I
Design future state onboarding process & intake spec	R	A	C	C	C	I	I	I
Adjust vendor export format to match specification	C	I	I	C	A	I	R	I
Build summary extraction & dashboard data feeds (technical)	C	I	C	R/A	I	C	I	I
Generate & QA monthly summary report	C	I	R/A	I	I	I	I	I
Submit monthly report to Ministry of Health	I	A	R	I	I	I	I	I
Approve / sign off final deliverables	R	A	C	C	C	C	I	I

RACI Legend

R	Responsible — Does the work
A	Accountable — Owns the outcome, final approval
C	Consulted — Provides input before/during the work
I	Informed — Kept updated on progress or outcome

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Phase 4 of 7: Requirements Document

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Table of Contents

1. Purpose.....	2
2. Requirements Prioritization Key.....	2
3. Business Requirements.....	2
4. Functional Requirements.....	2
4.1 Vendor Intake & Summary Extraction.....	2
4.2 Vendor Detail Data & Service Dashboards.....	3
5. Non-Functional Requirements.....	3
6. Assumptions & Dependencies.....	3
7. Out of Scope.....	3
8. Traceability Note.....	4

Requirements Document

Vendor Data Onboarding & Integration Process — Prairie Telehealth Network (PTN)

1. Purpose

This document defines the business, functional, and non-functional requirements for standardizing how PTN onboards telehealth vendor data. It builds on the Project Brief (Phase 1) and the current state process map and stakeholder analysis (Phases 2-3), and will be used to drive the data mapping specification (Phase 5) and future state process design (Phase 6).

2. Requirements Prioritization Key

Requirements are prioritized using MoSCoW: Must have (critical to project success), Should have (important but not critical), Could have (desirable, may be deferred).

3. Business Requirements

Business requirements describe the high-level organizational needs this project must satisfy.

ID	Business Requirement	Priority
BR-01	PTN must have a standardized process for onboarding new telehealth vendors' data into its reporting pipeline.	Must
BR-02	The solution must reduce manual reconciliation effort for monthly Ministry reporting from ~6-8 days to ~2 days per cycle.	Must
BR-03	PTN must be able to produce accurate, consistent monthly performance reports to the Ministry of Health across all vendors.	Must
BR-04	Internal program teams (Virtual Visits, Telehealth, RPM) must have access to vendor-level detail data to support service-specific dashboards.	Should
BR-05	The onboarding process must be repeatable and scalable to accommodate 2-3 new vendors within 12 months without additional headcount.	Must

4. Functional Requirements

4.1 Vendor Intake & Summary Extraction

These requirements cover how vendor detail data is standardized into a consistent monthly summary for Ministry of Health and cross-organizational reporting.

ID	Requirement	Priority	Linked BR
FR-01	Define a standardized intake specification (file format, delivery method, frequency) that all vendors must follow when submitting detailed usage and clinical data.	Must	BR-01
FR-02	Map each vendor's detailed data fields to a standardized summary schema containing common program metrics (e.g., total sessions, unique patients served, average session duration, SLA compliance rate).	Must	BR-01, BR-03
FR-03	Convert vendor-specific units (e.g., session duration in seconds vs. minutes) to a consistent standard unit defined in the summary schema.	Must	BR-03
FR-04	Generate a standardized monthly summary report suitable for submission to the Ministry of Health, aggregating data across all onboarded vendors.	Must	BR-02, BR-03
FR-05	Flag records that cannot be mapped to the summary schema (e.g., missing required fields) for manual review before report generation.	Should	BR-03

4.2 Vendor Detail Data & Service Dashboards

These requirements cover how each vendor's detailed data is retained and structured to support internal, service-specific dashboards.

ID	Requirement	Priority	Linked BR
FR-06	Retain each vendor's detailed dataset, in its native structure, for use in service-specific dashboards (Virtual Visits, Telehealth, RPM).	Should	BR-04
FR-07	Define which detail-level fields from each vendor are required inputs for each service-specific dashboard.	Should	BR-04
FR-08	Onboard new vendors using a documented, repeatable mapping template rather than ad hoc, one-off analysis.	Must	BR-05

5. Non-Functional Requirements

Non-functional requirements describe the qualities the solution must exhibit, independent of specific features.

ID	Requirement	Priority	Linked BR
NFR-01	Usability: Intake specification documentation must be clear enough for vendors with varying technical sophistication to implement without extensive back-and-forth clarification.	Must	BR-05
NFR-02	Performance: The summary extraction process must complete within 2 business days of receiving all vendor files for a reporting cycle.	Must	BR-02
NFR-03	Scalability: The data mapping approach must support onboarding at least 2 additional vendors per year without redesigning the summary schema.	Should	BR-05
NFR-04	Data Quality: The summary schema must enforce standard units, formats, and field definitions to eliminate discrepancies caused by vendor-specific conventions.	Must	BR-03
NFR-05	Security & Compliance: All vendor data, detail and summary, must be handled in accordance with PTN's data governance and provincial health information privacy standards.	Must	BR-01
NFR-06	Maintainability: The mapping specification and summary schema must be documented so PTN staff can update them without vendor involvement when internal reporting needs change.	Should	BR-05

6. Assumptions & Dependencies

- The summary schema and mapping specification will be defined jointly by the Business Analyst and Analytics & Reporting Team Lead (see Phase 5: Data Mapping).
- IT/Data Integration will implement the technical extraction logic based on the mapping specification; implementation itself is outside this project's scope.
- Vendors are assumed willing and able to adjust export processes if given sufficient lead time and clear documentation.

7. Out of Scope

- Actual ETL/system build or dashboard development.
- Real-time or event-driven data integration (batch/periodic submission only).
- Vendor contractual or legal terms.

8. Traceability Note

Each functional and non-functional requirement is mapped back to a Business Requirement (BR) to ensure every proposed feature or quality has a clear organizational justification. This traceability will carry forward into the Data Mapping Table (Phase 5), where each summary field will in turn be traced to the functional requirement it fulfils.

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Phase 5 of 7: Data Mapping Table

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Standardized Summary Schema

Unified monthly reporting schema target for Ministry of Health and cross-vendor reporting

Field Name	Data Type	Description	Example Value
Reporting_Month	Date (YYYY-MM)	Calendar month the summary covers	2026-06
Vendor_Name	Text	Name of the reporting vendor	TeleCare Connect
Service_Line	Text	Program the vendor supports: Virtual Visits, Telehealth, or RPM	Telehealth
Total_Service_Events	Integer	Total sessions / encounters / readings for the month (definition varies by service line — see mapping notes)	1,284
Unique_Patients_Served	Integer	Distinct patients served in the month	612
Avg_Duration_Minutes	Decimal (nullable)	Average session/encounter length in minutes; not applicable to reading-based services (e.g., RPM)	18.4
SLA_Compliance_Rate_Pct	Decimal (%)	Percentage of events meeting the defined service-level target	94.2%
Issues_Reported_Count	Integer	Count of flagged technical issues or clinical alerts (semantics vary by vendor — see mapping notes)	7
Data_Quality_Pct	Decimal (%) (nullable)	Percentage of records successfully transmitted/complete; primarily relevant for device-based data	98.6%

Note: Not every field applies identically across vendors/service lines. This intentional schema captures a common reporting core, while the Data Mapping sheet documents where a field is not applicable and why.

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Phase 5 of 7: Data Mapping Table
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Vendor Detail Schemas (Source)

Native data structure submitted by each fictional vendor, prior to mapping

Vendor	Source Field	Data Type	Description
MediLink Virtual Care	session_id	Text (unique ID)	Unique identifier for each virtual visit session
MediLink Virtual Care	patient_id	Text	Internal patient identifier used by MediLink
MediLink Virtual Care	provider_id	Text	Identifier for the clinician conducting the session
MediLink Virtual Care	session_start	Datetime	Session start timestamp
MediLink Virtual Care	session_end	Datetime	Session end timestamp
MediLink Virtual Care	duration_sec	Integer (seconds)	Session length in seconds

MediLink Virtual Care	visit_type	Text	Type of virtual visit (e.g., follow-up, initial consult)
MediLink Virtual Care	connection_quality_score	Integer (1-5)	Vendor's self-reported connection quality rating
MediLink Virtual Care	issue_flag	Boolean	True if a technical issue was logged during the session
TeleCare Connect	EncounterID	Text (unique ID)	Unique identifier for each telehealth encounter
TeleCare Connect	PatientRef	Text	Patient reference number used by TeleCare Connect
TeleCare Connect	ClinicianRef	Text	Reference number for the consulting clinician
TeleCare Connect	EncounterDate	Date	Date the encounter took place
TeleCare Connect	DurationMinutes	Integer (minutes)	Encounter length in minutes
TeleCare Connect	ConsultType	Text	Type of consultation (e.g., specialist referral, routine)
TeleCare Connect	SLAFlag	Text (Y/N)	Indicates whether the encounter met TeleCare's internal SLA target

TeleCare Connect	TechnicalIssueNotes	Text (nullable)	Free-text notes on technical issues; blank if none occurred
RPM Health Solutions	DeviceID	Text (unique ID)	Identifier for the remote monitoring device
RPM Health Solutions	PatientMRN	Text	Patient medical record number associated with the device
RPM Health Solutions	ReadingTimestamp	Datetime	Timestamp of the individual device reading
RPM Health Solutions	ReadingType	Text	Type of reading captured (e.g., blood pressure, glucose)
RPM Health Solutions	ReadingValue	Decimal	Recorded value of the reading
RPM Health Solutions	AlertTriggered	Boolean	True if the reading triggered a clinical alert threshold
RPM Health Solutions	TransmissionStatus	Text (Success/Failed)	Whether the reading was successfully transmitted from the device

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Phase 5 of 7: Data Mapping Table

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Data Mapping: Vendor Detail — Standardized Summary

Field-level transformation rules for each vendor, with exceptions noted

Target Summary Field	Vendor	Source Field(s)	Transformation Rule	Notes / Exceptions
Reporting_Month	MediLink Virtual Care	session_start	FORMAT(session_start , "MM-YYYY")	Extracting Month and year from session_start field.
Vendor_Name	MediLink Virtual Care	N/A — static	Hard-coded / assigned by the intake pipeline based on which vendor's file is being processed	Not present in vendor's raw data
Service_Line	MediLink Virtual Care	N/A — static	Assigned via a vendor-to-service-line reference lookup (MediLink → Virtual Visits)	Maintain this lookup table as new vendors are onboarded (see FR-08)
Total_Service_Events	MediLink Virtual Care	session_id	COUNT(DISTINCT session_id) per calendar month	Standard count of completed sessions
Unique_Patients_Served	MediLink Virtual Care	patient_id	COUNT(DISTINCT patient_id)	
Avg_Duration_Minutes	MediLink Virtual Care	duration_sec	AVG(duration_sec) / 60	Convert seconds to minutes

SLA_Compliance_Rate_Pct	MediLink Virtual Care	connection_quality_score	% of sessions with connection_quality_score >= 4	No native SLA field — quality score used as agreed proxy; confirm threshold with Analytics Lead
Issues_Reported_Count	MediLink Virtual Care	issue_flag	COUNT where issue_flag = TRUE	Technical issues only
Data_Quality_Pct	MediLink Virtual Care	N/A	N/A	No transmission/completeness field in source data; leave null for this service line
Reporting_Month	RPM Health Solutions	ReadingTimestamp	Derived from the month portion of ReadingTimestamp, aggregated across all readings in that file	
Vendor_Name	RPM Health Solutions	N/A — static	Hard-coded / assigned by the intake pipeline based on which vendor's file is being processed	Not present in vendor's raw data
Service_Line	RPM Health Solutions	N/A — static	Assigned via a vendor-to-service-line reference lookup (RPM Health Solutions – RPM)	Maintain this lookup table as new vendors are onboarded (see FR-08)
Total_Service_Events	RPM Health Solutions	ReadingTimestamp + DeviceID	COUNT of individual readings transmitted per calendar month	Semantic difference: this counts device readings, not sessions — call out in report footnotes
Unique_Patients_Served	RPM Health Solutions	PatientMRN	COUNT(DISTINCT PatientMRN)	
Avg_Duration_Minutes	RPM Health Solutions	N/A	N/A	RPM readings have no session duration concept; leave field null for this service line
SLA_Compliance_Rate_Pct	RPM Health Solutions	TransmissionStatus	% of readings where TransmissionStatus = 'Success'	Used here as the closest available proxy for service reliability

Issues_Reported_Count	RPM Health Solutions	AlertTriggered	COUNT where AlertTriggered = TRUE	Semantic difference: counts clinical alerts, not technical issues — must be labeled separately in the report, not blended with the other two vendors
Data_Quality_Pct	RPM Health Solutions	TransmissionStatus	% of readings where TransmissionStatus = 'Success'	Primary intended use of this field; device-based data makes transmission completeness meaningful here
Reporting_Month	TeleCare Connect	EncounterDate	FORMAT(EncounterDate , "MM-YYYY")	Extracting Month and year from EncounterDate field
Vendor_Name	TeleCare Connect	N/A — static	Hard-coded / assigned by the intake pipeline based on which vendor's file is being processed	Not present in vendor's raw data
Service_Line	TeleCare Connect	N/A — static	Assigned via a vendor-to-service-line reference lookup (TeleCare Connect – Telehealth)	Maintain this lookup table as new vendors are onboarded (see FR-08)
Total_Service_Events	TeleCare Connect	EncounterID	COUNT(DISTINCT EncounterID) per calendar month	Standard count of completed encounters
Unique_Patients_Served	TeleCare Connect	PatientRef	COUNT(DISTINCT PatientRef)	
Avg_Duration_Minutes	TeleCare Connect	DurationMinutes	AVG(DurationMinutes)	Already in minutes — direct pass-through
SLA_Compliance_Rate_Pct	TeleCare Connect	SLAFlag	% of encounters where SLAFlag = 'Y'	Direct pass-through calculation
Issues_Reported_Count	TeleCare Connect	TechnicalIssueNotes	COUNT where TechnicalIssueNotes is not blank	Technical issues only

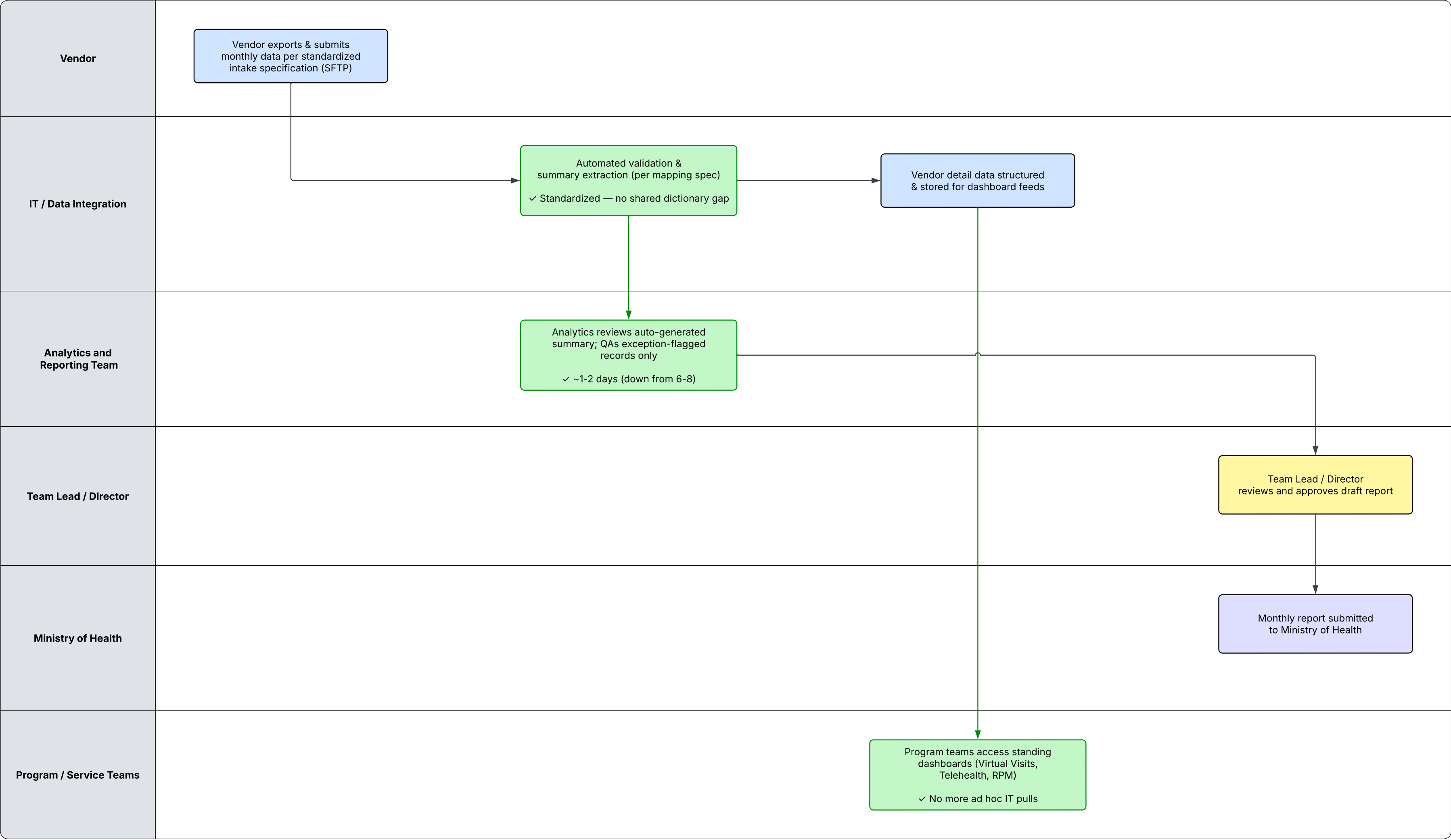
Data_Quality_Pct	TeleCare Connect	N/A	N/A	No transmission/completeness field in source data; leave null for this service line
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Note: Grey rows = field not applicable for that vendor/service line. This is a deliberate, documented exception not a data gap and should be preserved as null rather than defaulted to zero in the summary output.

Future State: Standardized Vendor Data Onboarding Process

Phase 6 - Automated summary extraction + standing service dashboards

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 Resolves a current-state pain point